



M. Paola Vaccaro

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Interests & Expertise

- Main interests:
 - Computational astrophysics
 - Gravitational-wave astronomy
 - Black hole dynamics
 - Stellar clusters
 - Stellar evolution
- Data analysis and visualization
- Scientific software development
- Methods:
 - Semi-analytical models
 - Hydrodynamics
 - Population synthesis
 - Cosmological simulations analysis
 - Bayesian statistics
- Numerical tools:
 - Python*
 - C/C++
 - Fortran
 - Linux
 - MPI
 - CUDA
 - Slurm*
 - HPC workflows*(*** = proficient)
- Scientific writing in LaTeX
- Conference presentations
- Science communication and outreach
- Collaborative research
- Mentoring
- Project management
- Languages:
 - Italian (mother tongue)
 - English (fluent, level C1)
 - Spanish (basic, level A2)
 - German (basic, level A1)

Education

Sep. 2023 – 2027 PhD in Astronomy and Cosmic Physics

At Institute for Theoretical Astrophysics (ITA), ZAH,
Heidelberg University in Heidelberg, Germany.

Supervisor: Prof. Michela Mapelli - Thesis title: “Binary Black Hole population in the Local Universe”

Oct. 2020 – Sep. 2022 Master degree in Astrophysics

At Università degli Studi di Padova in Padua, Italy.

Final grade: 110/110 *cum laude*, GPA: 29.4/30

Supervisor: Prof. Michela Mapelli - Thesis title: “Hierarchical Black Holes Mergers in Active Galactic Nuclei”

Sep. 2019 – Jul. 2020 Master courses in Matter Physics

At Università di Pisa in Pisa, Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 Bachelor degree in Physics

At Università di Pisa in Pisa, Italy.

Final grade: 110/110 *cum laude*, GPA: 28.6/30

Supervisor: Prof. Pier Giorgio Prada Moroni - Thesis title: “Cooling evolution of white dwarfs and cosmochronology”

Sep. 2011 – Jul. 2016 High school diploma

At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy.

Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 High school exchange year

At Mead High School in Mead, Colorado, USA.

GPA: 4.0/4 - ACT score: 33/36

Scientific publications *as main author*

- (IV) **AGN-driven BBH mergers: Black hole populations and hierarchical growth across the AGN parameter space**, [arxiv:2606.10823](#)
- (III) **The role of migration traps in the formation of binary black holes in AGN disks**, [A&A, 708, A171 \(2026\)](#)
- (II) **Astrophysics: a modern discipline with a Newtonian origin**, [Phil. Trans. R. Soc. A, 383: 20230294. \(2025\)](#); Invited contribution to a theme issue.
- (I) **Impact of gas hardening on hierarchical black hole mergers in AGN disks**, [A&A, 685, A51 \(2024\)](#)

Scientific publications

as co-author

- **The Science of the Einstein Telescope**, Adrian Abac et al. (incl. MPV), 2025, arXiv:2503.12263
- **The boring history of Gaia BH3 from isolated binary evolution**, Giuliano Iorio et al. (incl. MPV), A&A, 690, A144 (2024)
- **Hierarchical binary black hole mergers in globular clusters: mass function and evolution with redshift**, Stefano Torniamenti et al. (incl. MPV), A&A, 688, A148 (2024)
- **GW190521: a binary black hole merger inside an active galactic nucleus?**, Sophia Morton et al. (incl. MPV) 2023, Phys. Rev. D, 108, 123039
- **The hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology**, Jonathan R. Gair et al. (incl. MPV), 2022, AJ, 166, 22

Funding and affiliations

SSC Fellowship (2026 – 2027) Selected for individual mentoring and training in research software engineering and sustainable development at Scientific Software Center, Heidelberg University.

Membership of IMPRS (October 1st, 2023 – Present) PhD fellow of the 19th generation in the International Max Planck Research School (IMPRS) for Astronomy and Cosmic Physics at Heidelberg University.

Membership of the Einstein Telescope collaboration (October 1st, 2023 – Present), Division 3

Membership of the Young Research Convent (October 1st, 2023 – Present) of the Heidelberg Cluster of Excellence STRUCTURES. Program funded by the German Excellence Strategy fund.

Travel grants: Research visit to the GSSI in L'Aquila, Italy awarded through the ACME Transnational Access programme (April 2026); Participation in the IAU Symposium 398 and MODEST-25 hosted in Seoul, South Korea, sponsored by the International Astronomical Union (July 2025); Participation in the International HPC Summer School hosted in Kobe, Japan, sponsored by the EuroHPC Joint Undertaking via the HPC SPECTRA project (July 2024).

Pre-doctoral fellowship (February 1st – August 31st, 2023) in DEMOBLACK ERC Team at Università degli Studi di Padova, Padua, Italy. Supervisor: Prof. Michela Mapelli.

Internship in GW group (October 1st – December 31st, 2022) at IFAE-UAB, Barcelona, Spain. Supervisors: Dr. Lluïsa-Maria Mir i Martínez and Prof. Mario Martínez Pérez

Teaching experience and academic service

- Reviewer for *Astronomy & Astrophysics* (A&A) since 2025
- Reviewer for *Journal of Cosmology and Astroparticle Physics* (JCAP) since 2025
- Reviewer for *Physical Review D* (PRD) since 2026
- Teaching assistant for “Introduction to Astronomy and Astrophysics”: Summer Semesters 2024, 2025
Conducted daily exercise tutorials, corrected student submissions, and provided detailed feedback.
- Student co-supervision
 - Master student Julian L. Saling, “Hydrodynamical simulations of massive-star collisions” 2025/2026
 - Bachelor student Yannick Seif, “Pair Up of Binary Black Holes in Active Galactic Nuclei” 2024/2025
 - Bachelor student Jona A. Schubert, “The mass ratio of binary black holes: impact of mass transfer” 2024
- Volunteer staff member at EAS Annual Meetings: July 2024, July 2026
Supported registration and session logistics, including presentation slide checks and room coordination.

Conferences and workshops

- [Poster Presentation](#) | **EAS annual meeting** SwissTech Convention Centre, Lausanne, Switzerland
Session: “SS31: Exotic populations in star clusters seen through the lens of binary interactions”. Poster title: “Multi-Stage Collision Products in Dense Clusters”. 29 June–3 July, 2026
- [Contributed Talk](#) | **XVI Einstein Telescope Symposium** RWTH Aachen University, Aachen, Germany
Talk title: “Cosmic Evolution of Binary Black Hole Mergers from AGN Disks”. June 15th–19th, 2026
- [Contributed Talk](#) | **MODEST-25 & IAU Symposium 398** Seoul National University, Seoul, South Korea
Talk title: “Hierarchical Black Hole Mergers in AGN Disks”. June 16th–20th, 2025
- [Contributed Talk](#) | **Dynamix** Institute of Astronomy, Cambridge, UK
Title: “Hierarchical BH Mergers in AGN Disks: Tracing MBH Growth Across Cosmic Time”. June 2nd–6th, 2025
- [Poster Presentation](#) | **MODEST-24** Warsaw, Poland
(Remote attendance) August 19th–23rd, 2024
- [Invited Talk](#) | **New ideas on the origin of black hole mergers** Niels Bohr Institute, Copenhagen, Denmark
Talk title: “Specks of light from AGNs as possible indicators of the dynamical formation of binary black holes”. August 12th–16th, 2024
- [Contributed Talk](#) | **EAS annual meeting** Congress Center, Padova, Italy
Session: “S1: Unveiling black hole Growth across Cosmic Time in the JWST and LISA era”. July 1st–5th, 2024
Talk title: “Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks”.
- [Poster Presentation](#) | **EAS annual meeting** Congress Center, Padova, Italy
Session: “SS2: Spacetime ripples and light from compact binary mergers: Lessons learned from O4 so far, and a look at the future”. Poster title: “Specks of light from AGNs as possible indicators of dynamical formation of BBHs”. July 1st–5th, 2024
- [Contributed Talk](#) | **Massive Black Holes in the First Billion Years** University College Cork, Cork, Ireland
Talk title: “Possible constraints on the SMBH mass distribution from hierarchical BBH mergers in AGN discs”. April 29th – May 3rd, 2024
- [Contributed Talk](#) | **RESCEU-NBIA workshop on GW sources** University of Tokyo, Tokyo, Japan
Talk title: “The impact of Gas Hardening on Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks”. Contribution to the [discussion panel](#) “Many-Body Systems”. December 7th–15th, 2023
- [Flash Talk](#) | **Workshop "GwPopNext"** University of Milano Bicocca, Milan, Italy
Talk title: “Semi-analytical Models for Hierarchical Binary Black Hole Mergers”. July 10th–14th, 2023
- [Poster Presentation](#) | **Renaissance of Stellar Black-Hole Detections** Lorentz Center, Leiden, Netherlands
Invited participation June 26th–30th, 2023
- [Invited Talk](#) | **Physics of Data Workshop** Venetian institute of science, literature and art, Venice, Italy
Talk title: “How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes” April 19th–21st, 2023

Colloquia

- Department Colloquium GSSI, L’Aquila, Italy | April 24th, 2026
- Institute Seminar ISTA, Vienna, Austria | November 13th, 2025
- “BinBoom” Group Meeting (online) The University of Arizona | September 30th, 2025
- Physics of Stellar Objects Group Meeting HITS, Heidelberg, Germany | April 16th, 2025
- Gravitation and Cosmology Group Meeting GSSI, L’Aquila, Italy | December 9th, 2024
- ITP Cosmology Seminar ITP, Heidelberg, Germany | October 29th, 2024
- Einstein Telescope monthly meeting (online) ET-OSB Div3 | October 25th, 2024

- “Astroscopic” Seminar (online) GSSI, L’Aquila, Italy | April 26th, 2024
- Galaxy Group Meeting ITA-ZAH, Heidelberg, Germany | March 27th, 2024
- Planet Formation Group Meeting ITA-ZAH, Heidelberg, Germany | February 20th, 2024

Schools and workshops attendance

- “AI for Astronomy” Summer School hosted by the IWR in Heidelberg, Germany September 8th-12th, 2025
- International HPC Summer School hosted by the RIKEN Center in Kobe, Japan July 7th-12th, 2024
- “ML4ASTRO, Machine Learning in Astrophysics” workshop hosted by HITS in Heidelberg June 2024; July 2025
- “360° approach to Common Envelope Evolution: From Binary Progenitors to Remnants” hosted by Institut de Ciències del Cosmos of the Universitat de Barcelona (remote attendance) June 10th-14th, 2024
- 4th G2Net Training School “A network for Gravitational Waves, Geophysics and Machine Learning” hosted by Aristotle University of Thessaloniki (remote attendance) March 28th-30th, 2023
- Python Course (level advanced) hosted by ICFO in Barcelona, Spain November 24th-30th, 2022
- GW Open Data Workshop hosted by GWOSC (online event) May 23rd-25th, 2022
- IPAM Workshop III: Source inference and parameter estimation in Gravitational Wave Astronomy (remote attendance) November 15th-19th, 2021

Outreach and Media Coverage

- Research [featured in Gist.Science](#): “The formation of binary black holes in AGN disks”. April 8th, 2026
- Member of [Astronomers for Planet Earth](#) since 2025
- Author of “The Origins of Black Hole Mergers: Diverse Pathways Across the Universe”, [blog post](#) for the STRUCTURES Cluster of Excellence, aimed to a general audience. March 7th, 2025
- Co-organizer for the [workshop](#) “A dance of stars and black holes” of the Girl’s Day program “Physik hautnah!” for school-aged students 2024, 2025, 2026
- Co-author (with Dr. Giuliano Iorio and Dr. Erika Korb) of a pamphlet about stellar evolution, distributed to 12th-grade high schoolers in the school Liceo Volta (Reggio Calabria, Italy) by Scholé April 2024
- Speaker and staff member at the 2023 Scholé Summer School titled “PHYSIS”. Talks:
 - “Understanding Nature with Quantum Mechanics” with Prof. Giancarlo Cella (Pisa, Italy) July 25th, 2023
 - “The concept of space from Newton to Einstein” with Prof. Paolo Bussotti (Udine, Italy) July 29th, 2023
- Member of the organizing committee for the Scholé “School of Physics” for high schoolers since 2022
- Staff member of the cultural association [Scholé](#) based in Roccella Jonica (RC), Italy since 2020